Data & Methods draft: Fernhill Software, version 2 of 3 (the trial designer's model)

1. Unit of observation

One row of my data is one product team in one quarter, because the lot assigns whole teams and delivery is counted by the quarter. There are 1,440 rows, from 120 teams over 12 quarters, July 2023 to June 2026, eight before the pilot and four during it. The lot assigned teams, 40 to an arm, so arm comparisons rest on 40 teams each, not on 480 team-quarters.

2. Variable table

Concept	Variable	Kind	Source	What it misses
Structure and AI	`arm`	Labels	Pilot register	Drift from the arm
Before or after	`period`	Labels	Calendar	Half-done quarters
Unknowns	`escalations`	Counts	Tracker	Hallway answers
Waiting	`esc_wait_hrs`	Measurement (hours)	Tracker	Untracked waits
Difficulty	`hard_plan`	Yes/No	Planning review	Misjudged quarters
Delivery	`delivered_share`	Measurement (share)	Planning tool	Customer value and renegotiation

3. Data source

Fernhill's pilot register, IT licenses, issue tracker, planning review, and planning tool, read through People Analytics, cover all 120 teams for all 12 quarters. The planning review has rated difficulty only since July 2025. A second file from IT logs AI use by team for the four pilot quarters. The main file also holds records my model doesn't call for (escalations still open at quarter's close, departures, the lead's former role). Gemini suggested describing them, and I left them out. I use the usage logs only to check that the no-AI arm stayed off licensed tools. Fernhill is a useful fiction, so these records are simulated, but not from the model above. They come from a model this version doesn't see, whose units are these team-quarters and whose relationships, noise, and assumptions are published in Module 3. Until then, I treat them as the firm's records.

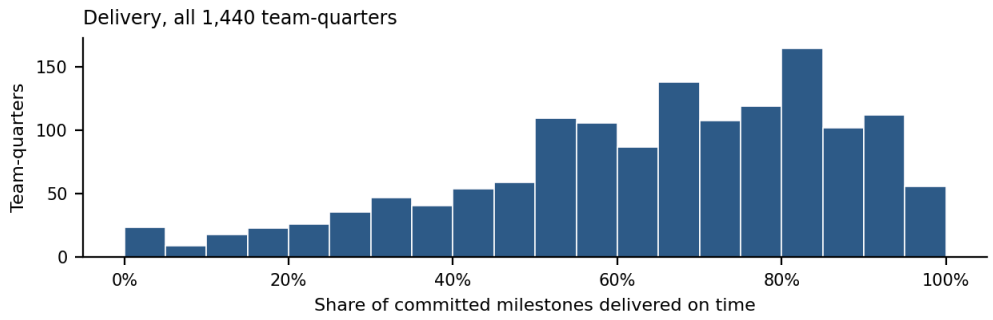
4. Quality concern

The difficulty rating at planning is the most likely to mislead. It is **mismeasured** whenever planners misjudge a quarter, and **missing** before July 2025, when nobody rated difficulty at planning. If planners misjudge, I would see quarters rated easy with as many escalations as hard ones, which makes the hard–easy difference look smaller. I will split results by the planning rating, never by the close-of-quarter rating made after the outcome, and say that misjudged quarters shrink any difference.

5. First description

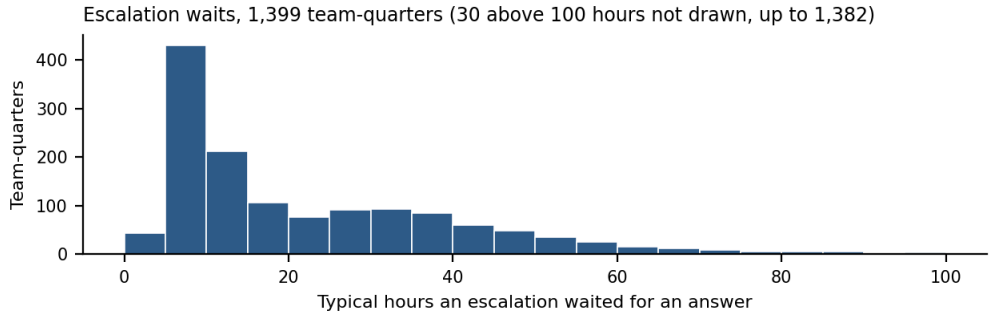
Delivery (`delivered_share`). *Kind:* a measurement, a share from 0 to 1. *How many:* 1,440, none blank. *Middle:* median 0.68, and mean 0.64, which a tail of bad quarters pulls down. *Spread:* half fall between 0.50

and 0.82, and the range is 0 to 1. *Shape:* skewed left, with a long tail of quarters that delivered little (24 under 5%). *Choice:* each team-quarter counts once, whatever its size. Weighting by milestones gives 0.63.



Against my model: not yet able to say, since one variable can't show whether the arms differ.

Waiting (`esc_wait_hrs`). *Kind:* a measurement, in hours. *How many:* 1,399, with 41 blank (3%). The gaps are in the tracker, before and during the pilot. *Middle:* median 15.5 hours, and mean 28.6, which a few very long waits pull up. *Spread:* half fall between 8.4 and 34.7 hours, and the range is 3.2 to 1,382. *Shape:* skewed far right, with 30 team-quarters waiting over 100 hours, the longest about eight weeks. *Choice:* blanks left out. Long waits kept, since nothing says they are errors, and the chart stops at 100 hours.



Against my model: consistent so far, since a long right tail is what a queue near capacity produces. Whose waits they are is a comparison for Module 2.

Difficulty (`hard_plan`). *Kind:* yes/No. *How many:* 480, all in the pilot, with 960 blank before July 2025, when the rating didn't exist. *Middle, spread and shape:* 0.44 rated hard (210 quarters). With two values, the proportion says it all. **Against my model:** not yet able to say, but there are enough quarters of each kind to compare the arms on hard and easy work.

Note on AI use

I used Gemini (Willamette account) for the counts, middles, spreads, and histograms. I kept the median for waits, since long waits pull the mean, and rejected its offer to drop waits over 100 hours, since nothing says they are errors.